Dynamic Partnership with Dissolution Risk

A Preliminary Analysis

September 15, 2026

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- **Marriage**, ending in divorce.
- **Professional partnership** — law, medical, consulting — ending when the partners split up.
- **R&D joint venture or alliance**, ending when it is terminated or a partner is acquired.

Also: start-up co-founders when one founder leaves; a currency or political union facing secession.

Model

First-Best Problem

Equilibrium

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- While together: joint flow $F(X_t)$, split equally $(F' > 0,\ F'' \leq 0)$
- The partnership ends at Poisson rate λ

Nothing is reversible: $\dot{X}_t = a_t + b_t \geq 0$.

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- Symmetric Nash bargaining:

$$u_A \;=\; d(A) + \tfrac{1}{2}\big\{S(X) - d(A) - d(B)\big\} \;=\; \tfrac{1}{2}S(X) + \tfrac{1}{2}\big(d(A) - d(B)\big).$$

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$u_A + u_B = S(X_\tau)$: the split is a pure transfer. So the planner below never sees d.

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$$\max_{n:\mathbb{R}_+\to[0,2]} \int_0^\infty e^{-(r+\lambda)t}\left[F(X_t)+\lambda S(X_t)-c\,n_t\right]\,dt,$$

subject to $\dot{X}_t = n_t$. (n_t denotes the joint investment $a_t + b_t$ at time t)

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Regularity Assumption

Define $M(X) := F'(X) + \lambda S'(X)$.

1. $M'(X) < 0$ for all $X \geq 0$;
   ($F(X) + \lambda S(X)$ is strictly concave; at least one of them is nonlinear)
2. $M(0^+) > (r+\lambda)c > \lim_{X\to\infty} M(X)$.

First-Best Characterization

Theorem

Under the regularity assumption, the optimal policy is bang-bang: invest at the maximum rate while $X_t < X^{FB}$ and stop thereafter, i.e.,

$$n_t^{FB} = \begin{cases} 2, & \text{if } X_t < X^{FB}, \\ 0, & \text{if } X_t > X^{FB}, \end{cases}$$

where

$$M(X^{FB}) = (r + \lambda)\, c.$$

HJB equation: Heuristic Derivation

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Over $[t, t+dt]$ the flow $F - cn$ accrues; with probability $\lambda\, dt$ the relationship *ends* and the pair collects the *lump* $S(X_t)$; otherwise it continues and is worth $W(X_{t+dt})$.

$$W(X_t) = \max_{n_t \in [0,2]} \Big\{ \big[F(X_t) - c\, n_t\big] dt + \underbrace{\lambda\, dt\, S(X_t)}_{\text{breakup}} + \underbrace{\big(1 - (r+\lambda)dt\big) W(X_{t+dt})}_{\text{survival}} \Big\} + o(dt)$$

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Using $W(X_{t+dt}) = W(X_t) + W'(X_t)\, n_t\, dt$, cancelling $W(X_t)$ and dividing by dt:

$$(r+\lambda)\, W(X) = \max_{n \in [0,2]} \big\{ F(X) + \lambda S(X) + n\big(W'(X) - c\big) \big\}$$

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$$W(X) = \frac{F(X) + \lambda S(X)}{r + \lambda} \qquad \Longrightarrow \qquad W'(X) = \frac{M(X)}{r + \lambda}.$$

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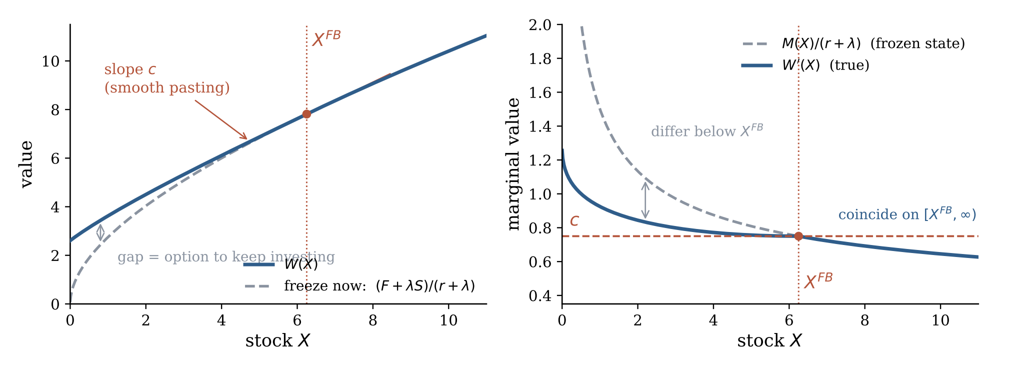
- Below X^{FB}, $M(X) > (r + \lambda)c$, so $W'(X) > c$ and $n = 2$. At the boundary $W'(X^{FB}) = c$: **smooth pasting**.

Optimal policy: the value function

$F(X) = \sqrt{X}$, $S(X) = X$; $r = 0.3$, $\lambda = 0.1$, $c = 0.75 \Rightarrow X^{FB} = 6.25$



Left: W lies above the freeze-now value below X^{FB} and coincides with it above, meeting tangentially with slope c. Right: the frozen-state slope is *not* W' below X^{FB}; they meet exactly at the threshold.

Equilibrium

Equilibrium: the concept

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- **Strategies.** Markov: a single function $\alpha(\text{own stock, other's stock}) \in [0,1]$, so

$$a_t = \alpha(A_t, B_t), \qquad b_t = \alpha(B_t, A_t).$$

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- **Equilibrium.** A *symmetric Markov perfect equilibrium* is an α that is a best response to itself.

Ava's problem

Ava takes α — Bob's strategy — as given and solves

$$\max_{a:\mathbb{R}_+\to[0,1]} \mathbb{E}\left[\int_0^\tau e^{-rt}\Big(\tfrac{1}{2}F(X_t) - c\,a_t\Big)dt \;+\; e^{-r\tau}\,u_A(A_\tau, B_\tau)\right],$$

subject to $\dot{A}_t = a_t,\ \dot{B}_t = \alpha(B_t, A_t),\ X_t = A_t + B_t$.

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The same Poisson step as in the planner's problem gives

$$\max_a \int_0^\infty e^{-(r+\lambda)t}\Big[\tfrac{1}{2}F(X_t){+}\lambda\,u_A(A_t,B_t){-}c\,a_t\Big]dt, \qquad u_A = \tfrac{1}{2}S(X){+}\tfrac{1}{2}\big(d(A){-}d(B)\big).$$

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Two changes from the planner: Ava gets $\frac{1}{2}F$ rather than F, and u_A rather than S.

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Using $V(A + a\,dt, B + b\,dt) = V + V_A\,a\,dt + V_B\,b\,dt$ and dividing by dt:

$$(r + \lambda)V = \max_{a\in[0,1]} \Big\{ \tfrac{1}{2}F(X) + \lambda u_A + \underbrace{b\,V_B}_{\text{Bob builds}} + a\big(V_A - c\big) \Big\}$$

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$$V_A(A, B) = \frac{1}{2}\frac{F'(X) + \lambda S'(X) + \lambda d'(A)}{r+\lambda} = \frac{1}{2}\frac{M(X) + \lambda d'(A)}{r+\lambda}.$$

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- On the symmetric path $A = B = X/2$, define the **private marginal value**

$$m(X) \;:=\; \frac{1}{2}M(X) + \frac{\lambda}{2}d'(X/2).$$

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Ava stops $\iff$ $m(X) \le (r+\lambda)c$. Compare $M(X) \le (r+\lambda)c$ for the planner.

Equilibrium Characterization

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Theorem

Under Regularity Assumption 2, in any symmetric Markov perfect equilibrium, accumulation stops at X^* given by

$$m(X^*) = (r+\lambda)\,c, \qquad m(X) = \tfrac{1}{2}M(X) + \tfrac{\lambda}{2}d'(X/2).$$

Comparison with the first best

Planner $M(X^{FB}) = (r+\lambda)c, \quad M(X) = F'(X) + \lambda S'(X)$

Equilibrium $m(X^*) = (r+\lambda)c, \quad m(X) = \frac{1}{2}M(X) + \frac{\lambda}{2}d'(X/2)$

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Theorem

$$X^* \lesseqgtr X^{FB} \qquad \Longleftrightarrow \qquad \lambda\, d'\big(X^{FB}/2\big) \lesseqgtr (r+\lambda)c$$

Proof: Draw a graph

Can the division rule restore efficiency?

Corollary (efficient division rule)

$X^* = X^{FB}$ if and only if

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Example

Let $S(X) = kX$. Then,

$$\frac{(r+\lambda)c}{\lambda} > k$$

But d cannot be that steep for free

Recall **feasibility**: $d(A) + d(B) \leq S(A + B)$. At the efficient symmetric split $A = B = X^{FB}/2$ this reads $2\,d(X^{FB}/2) \leq S(X^{FB})$.

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Proposition (efficiency requires losses in disagreement)

Let d be concave and feasible. If $X^* = X^{FB}$ then

$$d(0) \;\leq\; \frac{X^{FB}}{2}\left[\frac{S(X^{FB})}{X^{FB}} - \frac{(r+\lambda)c}{\lambda}\right].$$

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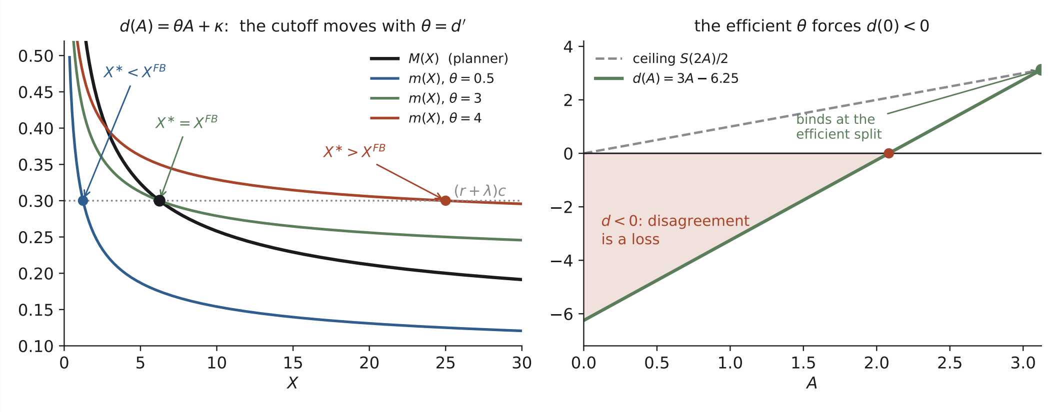
Whenever $S(X) = kX$, efficiency forces $d(0) < 0$: disagreement must be a loss.

Comparison: a numerical example

$F(X) = \sqrt{X}$, $S(X) = X$, $d(A) = \theta A + \kappa$; $r = 0.3$, $\lambda = 0.1$, $c = 0.75 \Rightarrow X^{FB} = 6.25$



Efficiency needs $\theta = (r+\lambda)c/\lambda = 3$, against a feasibility ceiling of $k = 1$ for proportional rules.

Binding feasibility at the efficient split gives $\kappa = \frac{1}{2}\left[S(X^{FB}) - \theta X^{FB}\right] = -6.25$: